

1: Introduction to Operating System (OS)

WHAT IS OPERATING SYSTEM?

Operating System (OS) ek **system software** hai jo:

- ✓ User aur Hardware ke beech interface provide karta hai
- ✓ Computer resources manage karta hai
- ✓ Applications ko run karne me help karta hai

Simple Definition

Operating System is a software that acts as an interface between user and computer hardware.

Real Life Example

Socho tum:

- Chrome use kar rahe ho
- Music chala rahe ho
- VS Code open hai
- Background me download chal raha hai

👉 In sabko manage kaun kar raha hai?

✓ Operating System

Examples of Operating Systems

OS	Devices
Windows	PC/Laptop
Linux	Servers

OS	Devices
Android	Mobile
macOS	Apple devices

MAIN FUNCTIONS OF OS

1 Process Management

Running programs ko manage karta hai.

Example:

Chrome + Spotify + VS Code ek sath chalna.

Work:

- Process create/delete
 - CPU allocate
 - Scheduling
-

2 Memory Management

RAM ko manage karta hai.

Work:

- Memory allocation
- Deallocation
- Track free memory

Example:

Game close karte hi RAM free ho jana.

3 File Management

Files aur folders handle karta hai.

Operations:

-  Create
 -  Delete
 -  Rename
 -  Read/Write
-

4 Device Management

Input/output devices control karta hai.

Examples:

- Keyboard
 - Mouse
 - Printer
 - Speaker
-

🔒 Security

System ko unauthorized access se protect karta hai.

Features:

- Password
 - Authentication
 - Access control
-

🧠 IMPORTANT TERMS

◆ Kernel

OS ka core part.

Responsible for:

- CPU management
- Memory management
- Device handling

Placement Tip:

“Kernel is the heart of Operating System.”

◆ System Call

User program → OS service request.

Examples:

open()
read()
write()

◆ Booting

Computer start hone ki process.

Types:

Type	Meaning
Cold Booting	Power ON from OFF state
Warm Booting	Restart

**TYPES OF OPERATING SYSTEM**

❏ Batch Operating System

Jobs batches me execute hoti hain.

Example:

Payroll systems

Disadvantage:

No direct interaction.

2 Multitasking OS

Ek user multiple tasks ek sath perform kar sakta hai.

Example:

Windows

3 Multiuser OS

Multiple users same system use karte hain.

Example:

Linux server

4 Real Time Operating System (RTOS)

Fixed time me response dena hota hai.

Examples:

- Air traffic control
- Robotics
- Medical systems

 GOALS OF OPERATING SYSTEM

-  Convenience
-  Efficiency
-  Resource Utilization
-  Performance

INTERVIEW / PLACEMENT QUESTIONS

Q1. What is Operating System?

 **Answer:**

Operating System is system software that acts as interface between user and hardware and manages computer resources.

Q2. What are the main functions of OS?

 **Answer:**

- Process management
- Memory management
- File management
- Device management
- Security

Q3. What is Kernel?

 **Answer:**

Kernel is the core component of OS responsible for resource management.

Q4. Difference between Application Software and OS?

OS	Application Software
System software	User software
Manages hardware	Performs specific task

Q5. What is multitasking?

 Answer:

Executing multiple tasks simultaneously.

Operating System interviews almost always start from:

- **Process**
- **Thread**
- **Scheduling**
- **Synchronization**

1 WHAT IS A PROCESS?

Formal Definition

A process is a program that is currently executing.

But placement interviews me sirf ye line enough nahi hoti.

Interviewer deeper understanding dekhna chahta hai.

Real Understanding

Jab tum koi application run karte ho:

- Memory allocate hoti hai
- CPU time milta hai
- Resources allocate hote hain
- Execution start hota hai

Tab OS us running entity ko:

 Process

kehta hai.

Example

Suppose tumne:

- Chrome open kiya
- VS Code open kiya
- Spotify chalaya

OS ke perspective se:

Application	Process
Chrome	Process
VS Code	Process
Spotify	Process

Har process ka:

- apna memory space
- apna execution state
- apne resources

hote hain.

Internal Structure of Process

Interviewers bahut baar puchte hain:

“What does a process contain internally?”

A process sirf code nahi hota.

Uske andar multiple memory sections hote hain.



1. Text Section (Code Section)

Program instructions store hoti hain.

Example:

```
cout<<"Hello";
```

Compiled machine instructions yaha store hoti hain.



2. Data Section

Global aur static variables.

Example:

```
int x = 10;
```



3. Stack Section

Function calls + local variables.

Example:

```
void fun(){  
    int a = 5;  
}
```

a stack me store hota hai.



4. Heap Section

Dynamic memory allocation.

Example:

```
new int[100];
```

Runtime memory allocation heap me hoti hai.

Interview Explanation

A good interview answer:

“A process contains code section, data section, stack and heap memory.”

2 PROCESS LIFE CYCLE

Every process execution ke during multiple states me move karta hai.

Main Process States

New → Ready → Running → Waiting → Terminated

◆ New State

Process create hua hai.

Example:

Tumne app icon click kiya.

OS process create kar raha hai.

◆ Ready State

Process memory me loaded hai but CPU ka wait kar raha hai.

Yani:

Ready to execute but not executing.

◆ Running State

CPU actively instructions execute kar raha hai.

◆ Waiting State

Process kisi external event ka wait kar raha hai.

Example:

- File read
 - Internet response
 - Database query
-

◆ Terminated State

Execution complete.

Example:

App close.

Real-World Example

Suppose YouTube app:

Action	State
Open app	New
Waiting for CPU	Ready
Playing video	Running
Buffering	Waiting
Close app	Terminated

Most Asked Interview Question

“Why does waiting state exist?”

Because CPU continuously execute nahi kar sakta.

Kabhi process:

- disk wait karta hai
- network wait karta hai
- input wait karta hai

Tab CPU dusre process ko de diya jata hai.

Isi se multitasking efficient hoti hai.

3 WHAT IS A THREAD?

Definition

Thread is the smallest unit of execution inside a process.

Simple words me:

Process ke andar execution path.

Real-World Understanding

Suppose Chrome browser:

Chrome ek process hai.

But Chrome ke andar:

- one tab
- another tab
- download manager
- video player

sab simultaneously run hote hain.

Ye sab:

 **Threads**

hote hain.

Important Concept

All threads:

- same memory share karte hain

- same resources use karte hain

Isliye:

Threads are lightweight.

Example

Suppose Word document:

Work	Thread
Typing	Thread 1
Spell check	Thread 2
Auto save	Thread 3

Sab simultaneously.

4 PROCESS VS THREAD

Most important placement question.

Feature	Process	Thread
Weight	Heavyweight	Lightweight
Memory	Separate	Shared
Resource Sharing	No	Yes
Switching Speed	Slow	Fast
Creation Cost	High	Low
Communication	Difficult	Easy

Deep Interview Understanding

Why process switching is slow?

Because:

- memory switch hoti hai
- resources reload hote hain
- PCB switch hota hai

High overhead.

Why thread switching is fast?

Because:

- same memory share hoti hai
 - fewer resources
 - less overhead
-

Best Interview Line

“Threads are faster because they share process resources.”

5 MULTITHREADING

Definition

Executing multiple threads concurrently.

Real-World Example

YouTube app me:

- video play

- comments load
 - ads loading
 - recommendations fetch
- sab simultaneously hote hain.
-

Why Multithreading Matters?

Because modern applications:

- responsive honi chahiye
 - fast honi chahiye
 - parallel work karna hota hai
-

Advantages

- ✓ Better responsiveness
 - ✓ Better CPU utilization
 - ✓ Faster execution
 - ✓ Parallelism
-

Disadvantages

- ✗ Synchronization problems
 - ✗ Race conditions
 - ✗ Debugging difficult
-

6 CONTEXT SWITCHING

CPU ek time par ek hi task execute karta hai.

But user ko lagta hai:

- Chrome
- VS Code
- Spotify

sab simultaneously chal rahe.

How?

Answer:

 Context Switching

CPU rapidly:

- process save karta hai
- dusra process load karta hai

Important

Process Context Switch

Slower

Thread Context Switch

Faster

Interview Tip

Agar interviewer puche:

“Why threads improve performance?”

Answer:

- lower switching overhead
 - shared memory
 - lightweight execution
-

7 PROCESS CONTROL BLOCK (PCB)

Every process ke liye OS ek data structure maintain karta hai:

 **PCB**

PCB Contains

Data	Purpose
Process ID	Unique identity
State	Running/Waiting
Registers	CPU execution info
Program Counter	Next instruction
Memory Info	RAM allocation

Real Understanding

PCB basically:

“Identity Card + Status Record”

of process.

Interview-Level Explanation

“PCB stores all information required to resume process execution.”

8 MPORTANT INTERVIEW QUESTIONS

? Difference between Program and Process?

Program = static file

Process = running instance of program

? Why are threads called lightweight?

Because they share memory/resources.

? Why is multithreading useful?

Improves responsiveness and performance.

? Why process switching is expensive?

Because memory and execution state must be switched.

? Why does browser use multiple threads?

For:

- parallel tasks
 - smooth UI
 - responsiveness
-

? What happens during context switching?

CPU saves current process state and loads another process state.



CPU Scheduling :



Sabse Pehle Samjho CPU Scheduling Kyu Chahiye?

Computer me ek time par:

- Chrome
- VS Code
- Spotify
- YouTube

sab chal rahe hote hain.

But:

CPU ek time par sirf ek process execute karta hai.

To OS ko decide karna padta hai:

“Ab CPU kis process ko diya jaye?”

Isi decision process ko:



CPU Scheduling

kehte hain.



Real Life Example

Suppose:

Ek teacher hai aur 5 students doubts puch rahe.

Teacher:

ek time par sirf ek student ko answer de sakta hai.

Teacher decide karega:

kaun pehle ayega,

kaun baad me.

Ye:

CPU Scheduling

jaisa hi hai.

Technical Definition

CPU Scheduling is the process of selecting which process gets CPU execution time.

Goals of CPU Scheduling

OS chahta hai:

- ☒ CPU idle na rahe
 - ☒ Processes fast complete ho
 - ☒ Waiting kam ho
 - ☒ Fairness ho
-

Important Terms

Placement interviews me directly puchte hain.

1 Arrival Time (AT)

Process kab system me aaya.

2 Burst Time (BT)

CPU ko kitna execution time chahiye.

3 Completion Time (CT)

Process kab complete hua.

4 Turnaround Time (TAT)

Total time from arrival to completion.

Formula

$$TAT = CT - ATT$$

Example

Arrival:

2 sec

Complete:

10 sec

Then:

$$TAT = 10 - 2 = 8 \text{ sec}$$

5 Waiting Time (WT)

CPU wait kitna kiya.

Formula

$$WT = TAT - BT$$

Example

TAT:

8 sec

Burst:

5 sec

Then:

$$WT = 8 - 5 = 3 \text{ sec}$$

Response Time

First response milne ka time.

Types of CPU Scheduling

Mainly 2 types:

Type	Meaning
Preemptive	CPU forcefully le sakte
Non-Preemptive	Once started, continue

Example

Non-Preemptive

Teacher ek student ka full doubt solve karega,
phir next.

Preemptive

Teacher बीच me interrupt karke:
urgent student ko priority de sakta hai.

FCFS (First Come First Serve)

Sabse simple scheduling algorithm.

Concept

Jo process pehle aaya:
vo pehle execute hoga.

Real Life Example

Ticket counter line.

Jo pehle line me:
vo pehle ticket.

Example

Process	Burst Time
P1	5
P2	3
P3	2

Execution:

P1 → P2 → P3

Advantages

-  Simple
 -  Easy implementation
-

Disadvantages

-  Long waiting time
 -  Convoy effect
-

Convoy Effect

Small processes long process ke piche wait karte hain.

Interview Tip

“FCFS is simple but inefficient for small processes.”

📌 SJF (Shortest Job First)

Most important algorithm.

🧠 Concept

Jiska burst time sabse kam:
vo pehle execute.

🌍 Real Life Example

Restaurant.

Small order pehle complete.

Example

Process	BT
P1	10
P2	2
P3	4

Execution:

P2 → P3 → P1

Main Advantage

✓ Minimum average waiting time.

Main Problem

✗ Burst time pehle se pata hona chahiye.

Interview Question

? Why SJF is optimal?

✓ Because it minimizes average waiting time.

ROUND ROBIN (RR)

Very important for placements.

🧠 Concept

Har process ko:
fixed time quantum milta hai.

🌍 Real Life Example

Teacher har student ko:
2 min deta hai.

Time khatam:
next student.

Example

Quantum:

2 ms

Processes:

Process	BT
P1	5
P2	4

Execution:

P1 → P2 → P1 → P2 → P1

Advantages

- ✓ Fairness
 - ✓ Good response time
 - ✓ Time-sharing systems
-

Disadvantages

- ✗ Too much context switching.
-

Important Interview Question

? What happens if time quantum too small?

- ✓ More context switching overhead.
-

? What if quantum too large?

- ✓ RR becomes FCFS.
-

4 PRIORITY SCHEDULING

Concept

Higher priority process pehle execute.

Example

Hospital emergency patients.

Serious patient first.

Problem

Starvation

Low priority process forever wait.

Solution

Aging

Old process ki priority increase kar do.

Interview Tip

“Aging prevents starvation.”

CONTEXT SWITCHING

CPU ek process se dusre me switch karta hai.

Example

CPU:

- Chrome
- VS Code
- Spotify

rapidly switch karta hai.

Problem

Context switching:
useful work nahi karta.

Extra overhead hota hai.

Interview Point

Too much context switching:
 performance decrease.

Throughput

Per unit time:
kitne processes complete hue.

Formula

Throughput = $\frac{\text{Number of completed processes}}{\text{Time}}$

Goal

Higher throughput better.

Comparison of Algorithms

Algorithm	Best Feature	Problem
FCFS	Simple	Long waiting
SJF	Minimum WT	Burst unknown
RR	Fairness	Context switching
Priority	Important tasks first	Starvation

MOST ASKED INTERVIEW QUESTIONS

What is CPU scheduling?

 CPU scheduling decides which process gets CPU.

Difference between preemptive and non-preemptive?

 Preemptive:
CPU can be taken away.

 Non-preemptive:
Process runs until completion/block.

Why Round Robin used in OS?

 Fairness and fast response.

Why SJF is optimal?

✓ Minimum average waiting time.

? What is starvation?

✓ Process waits forever.

? How to prevent starvation?

✓ Aging.

? What is convoy effect?

✓ Small processes wait behind large process in FCFS.

? What happens if time quantum too small?

✓ More context switching.

MEMORY TRICKS

FCFS

“First Come → First Serve”

SJF

“Small Work First”

RR

“Everyone gets equal turn”

Priority

“Important task first”
